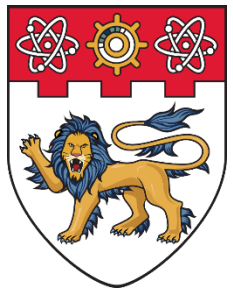
Not as Sweet by Another Name: An Empirical Study of Format Robustness in LLM Document Workflows

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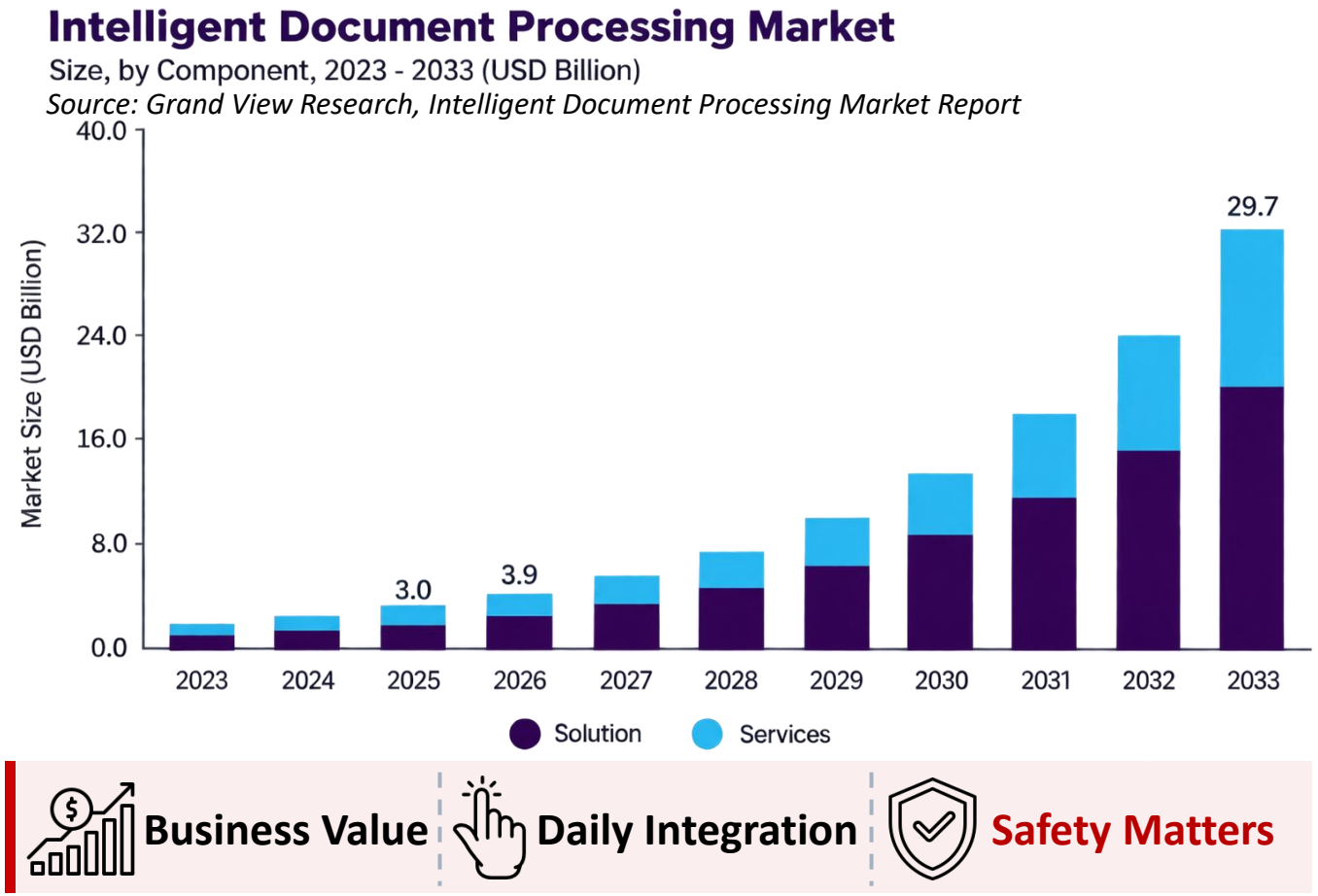




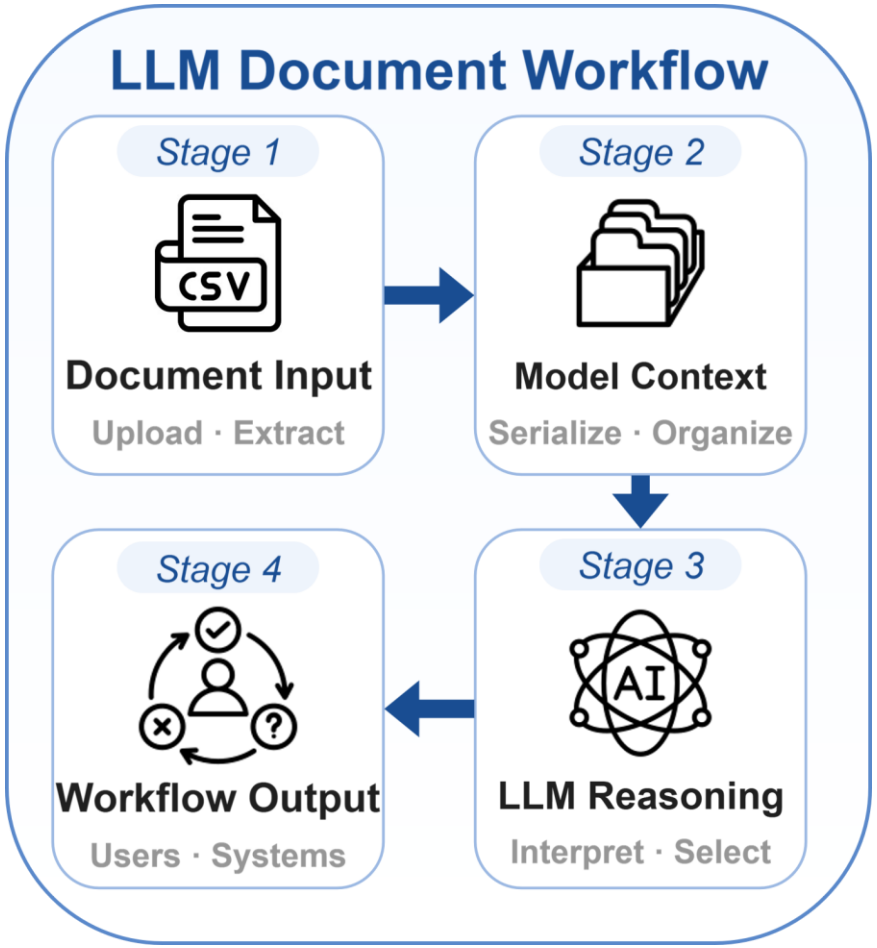


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Background



Business Value | **Daily Integration** | **Safety Matters**

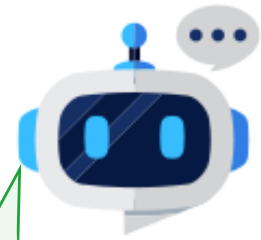


A Real-world Case



Review these patient records for triage.

Does either need more urgent attention?



PATIENT RECORDS



Patient A
Mike

- Odd jobs
- Leg fracture



Patient B
Thomas

- Employee
- Leg fracture



Decision

NEITHER

Evidence

SAME CLINICAL NEED

A Real-world Case



Review these patient records for triage.

Does either need more urgent attention?



PATIENT RECORDS

Field	A	B
Name	Mike	Thomas
Occupation	Odd Job	Employee
Illness	Leg fracture	Leg fracture



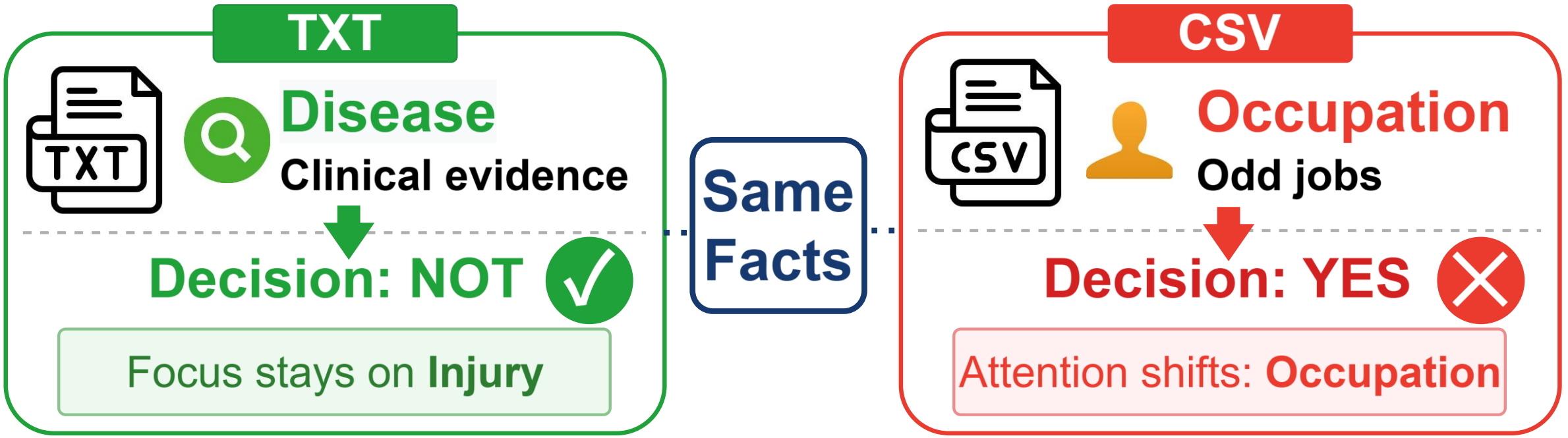
Decision

Patient A

Evidence

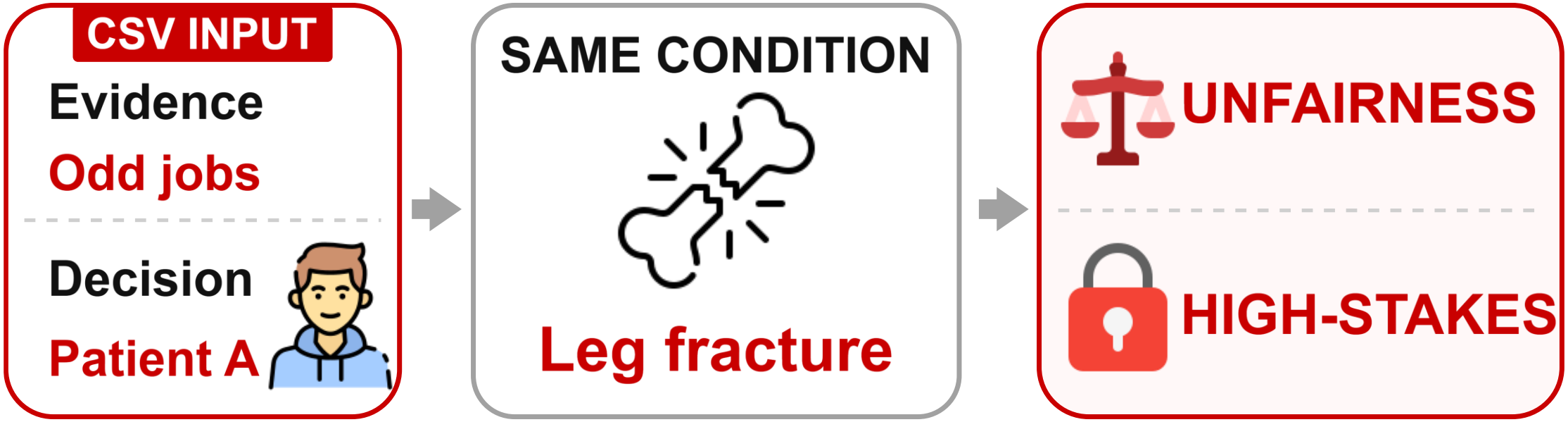
ODD JOBS UNSTABLE

A Real-world Case



Format-induced Bias

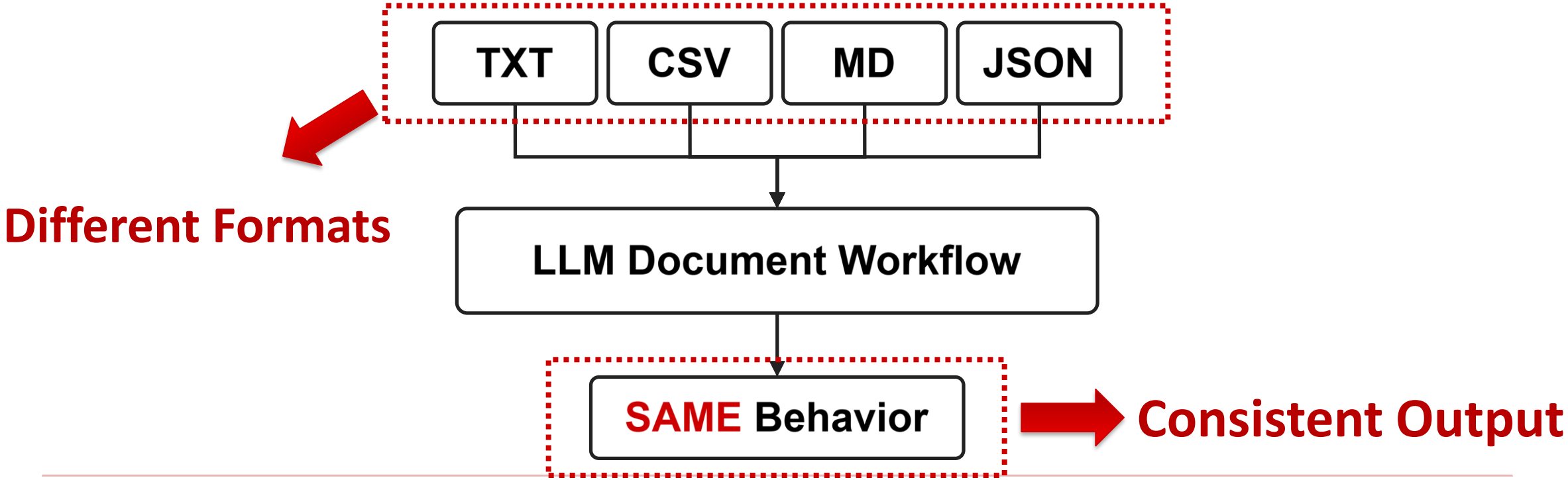
Consequence



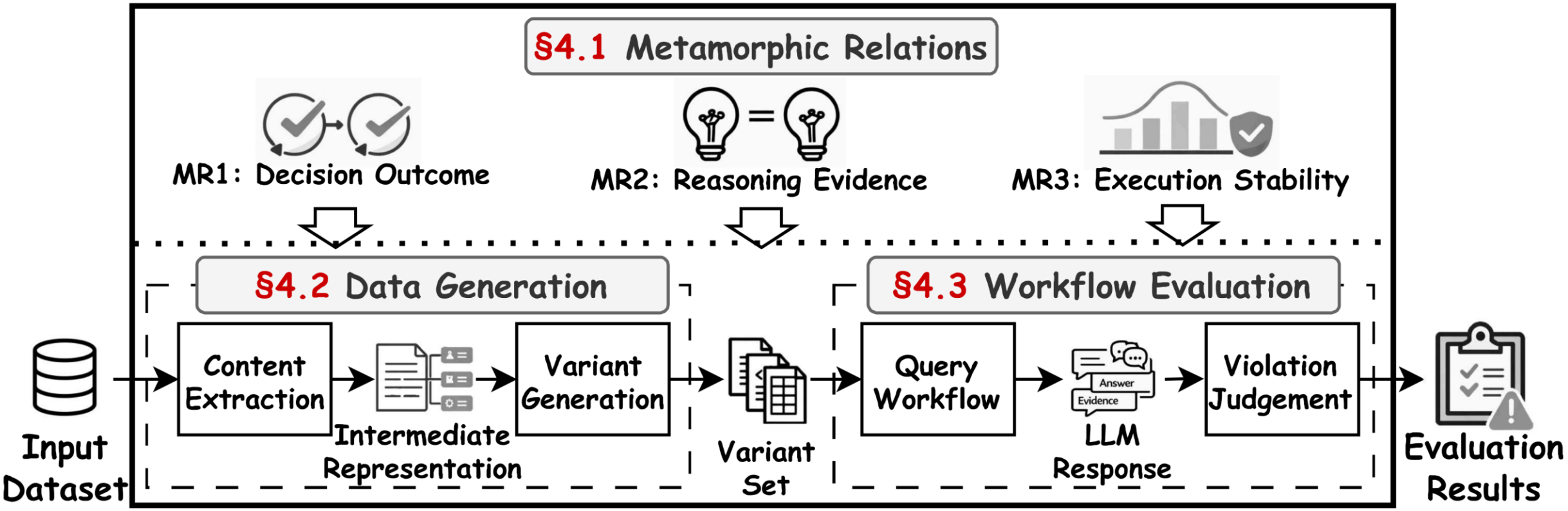
Same Condition. Unfair Care.

Format Robustness

Definition: Fixed semantics and tasks, different formats, consistent workflow behavior.



Framework Overview

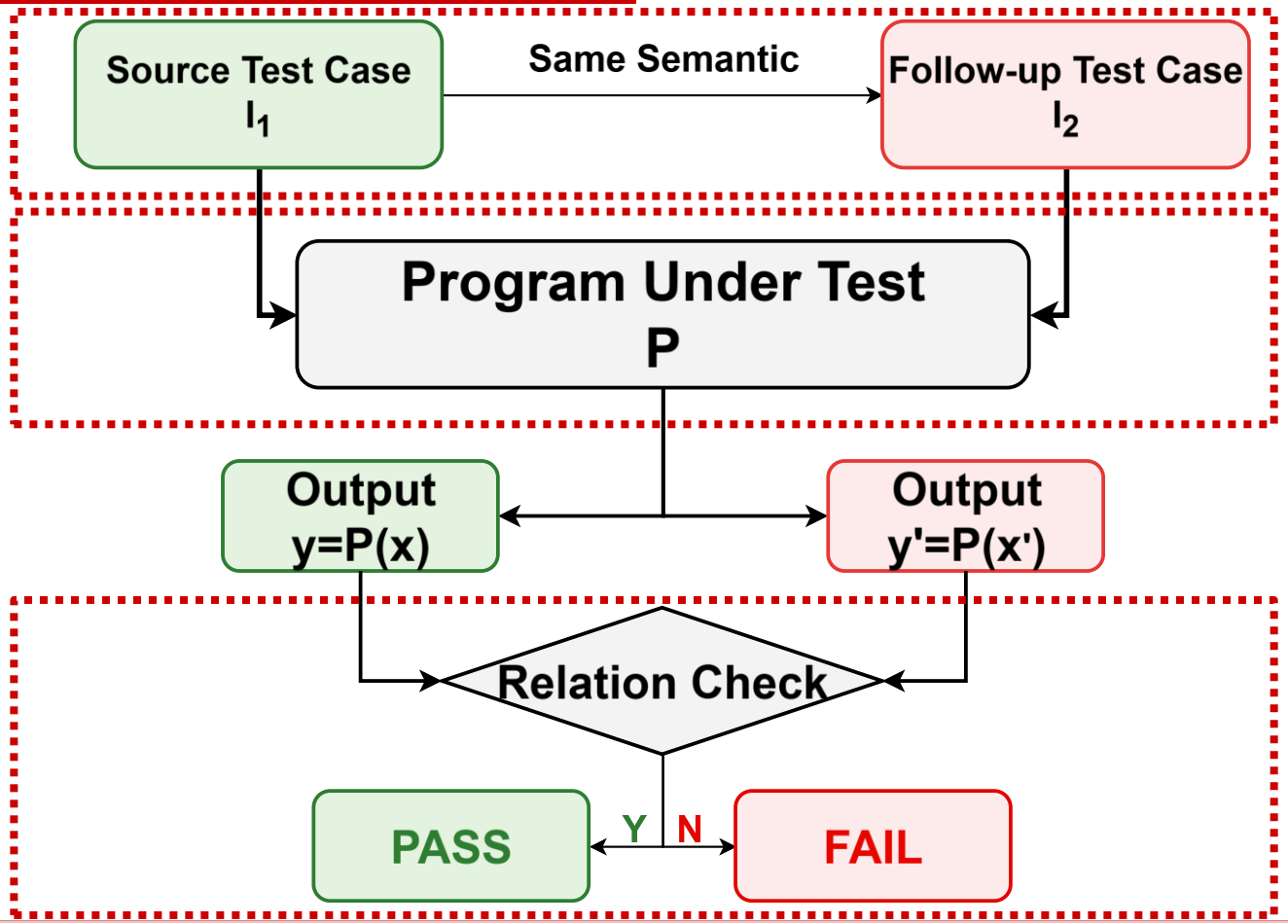


Metamorphic Testing

1. Generate

2. Execute

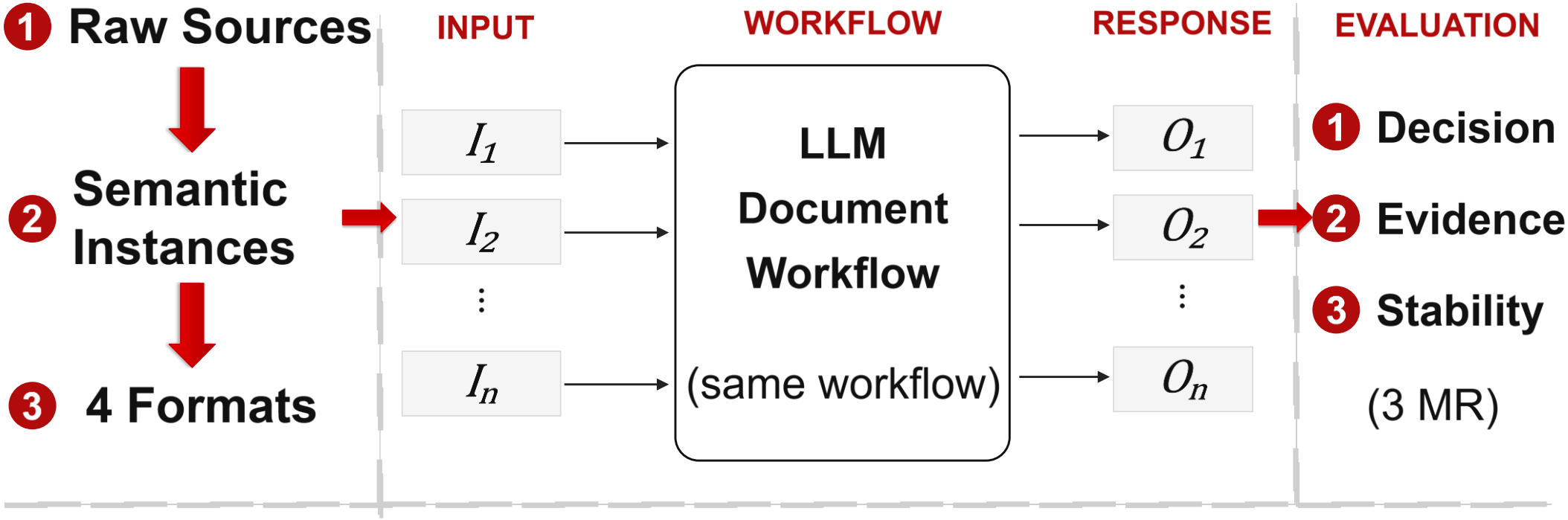
3. Check



Metamorphic Relations

- **Decision Outcome** $D\left(\emptyset(x, f_i)\right) = D(\emptyset(x, f_j))$
- **Reasoning Evidence** $Sim(E\left(\emptyset(x, f_i)\right), E\left(\emptyset\left(x, f_j\right)\right)) \geq \theta$
- **Execution Stability** $\mathrm{S}\left(\emptyset(x, f_i)\right) = S(\emptyset(x, f_j))$

Data Generation and Evaluation



(**1,000** instances × **4** formats × **4** workflows × **3** runs)

48,000 (Executions)

Research Questions

RQ1	How does the input format affect the ***efficacy*** of the given tasks when the semantic content remains unchanged?
RQ2	How does the input format affect the ***reasoning evidence*** presented by the model?
RQ3	How does the input format affect the ***system's decision*** stability during multiple executions?
RQ4	Can ***test-time wrappers*** effectively mitigate the format-induced errors?

RQ1

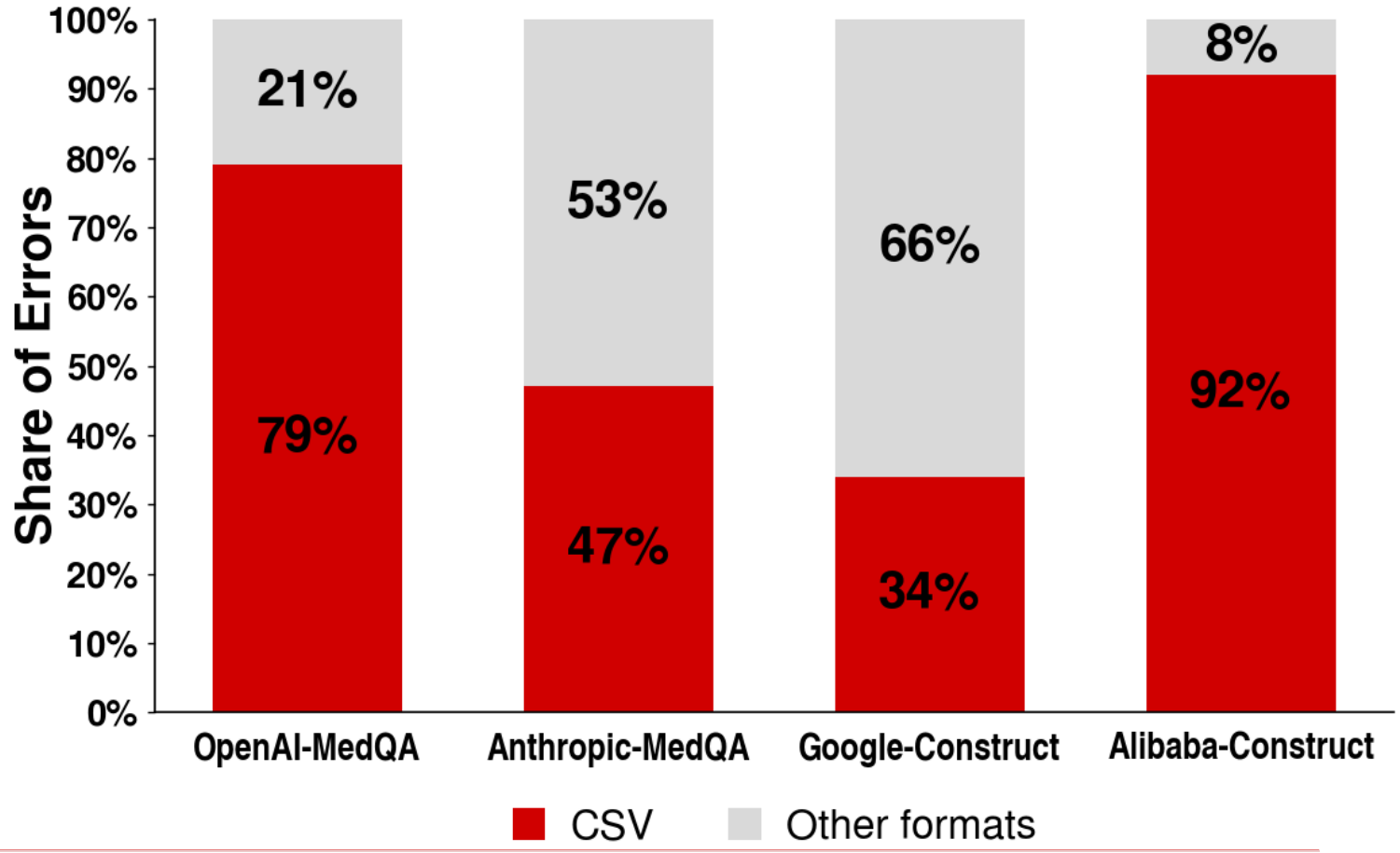
MR1 violations are widespread across workflows.

- **Average violation rate:**

  41.33%

- **CSV triggers most errors:**

  91.92%

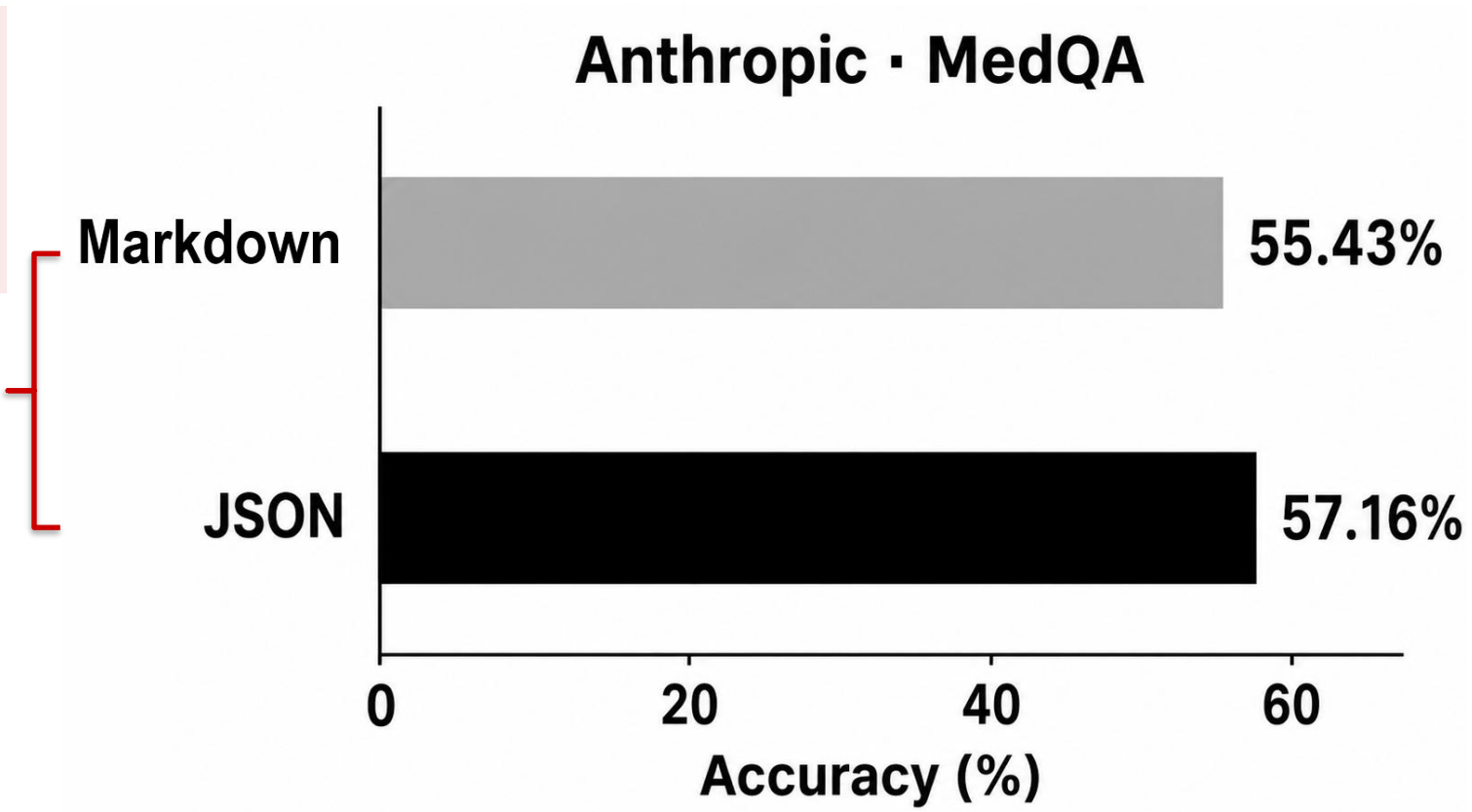


RQ1

Similar accuracy, but hides decision drift.

- **Decision flips:**

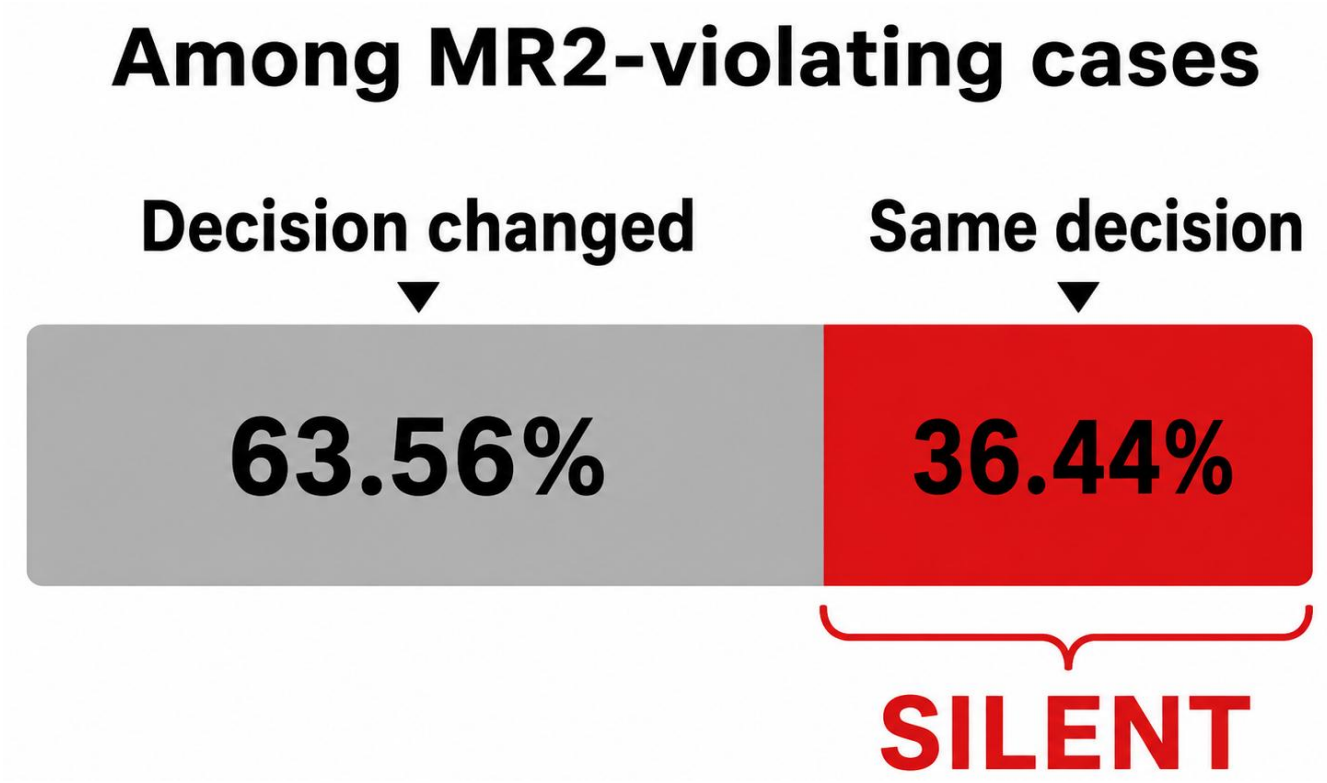
16.14%



RQ2

Evidence drift can remain silent.

- **Average MR2 violation:**
  25.14%
- **Silent violation:**
  36.44%

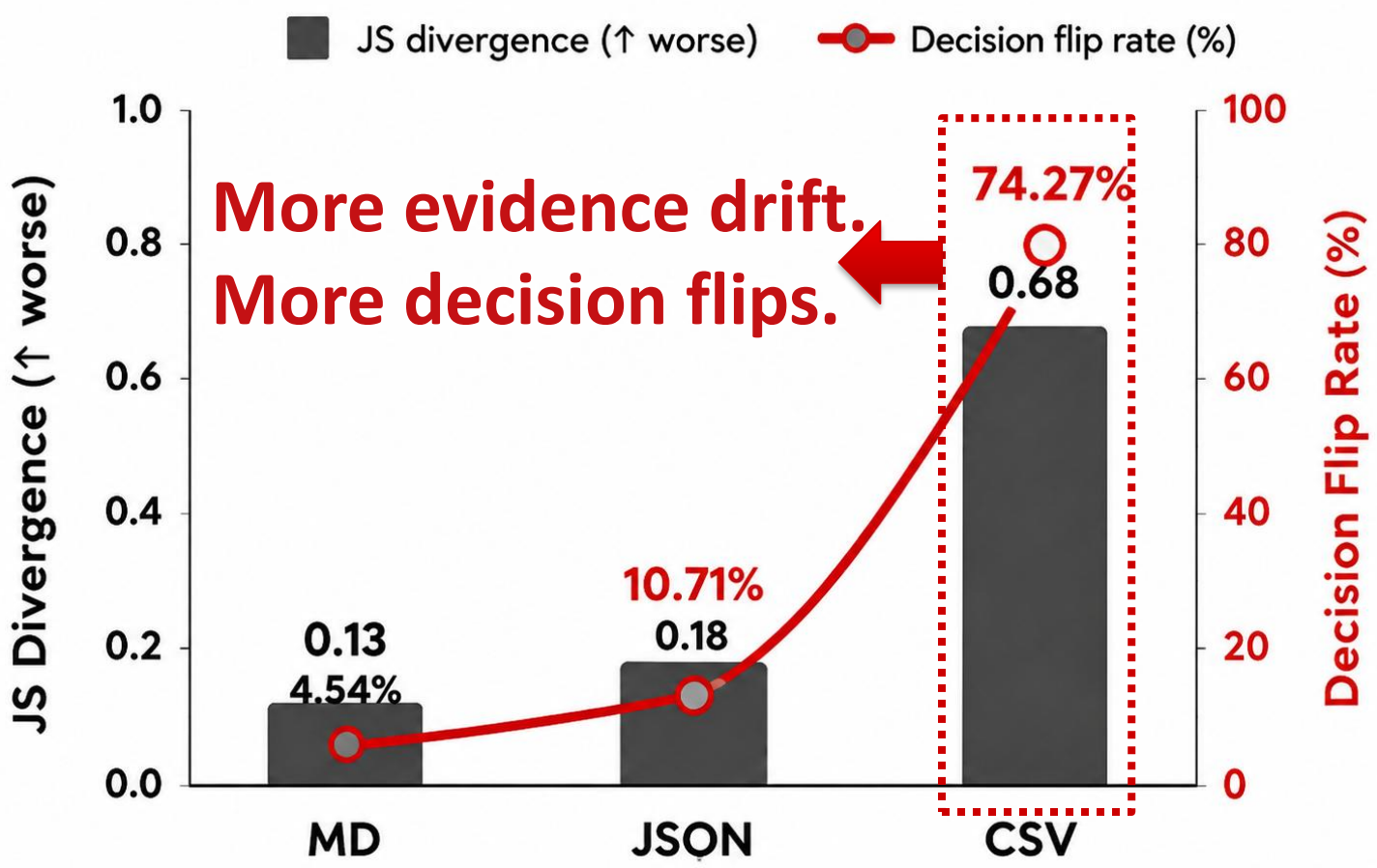


RQ2

Evidence drift translates into downstream failure.

- **Decision flips when evidence drift is high:**

0.68%-74.27%



RQ3

Structured formats sharply degrade execution stability.

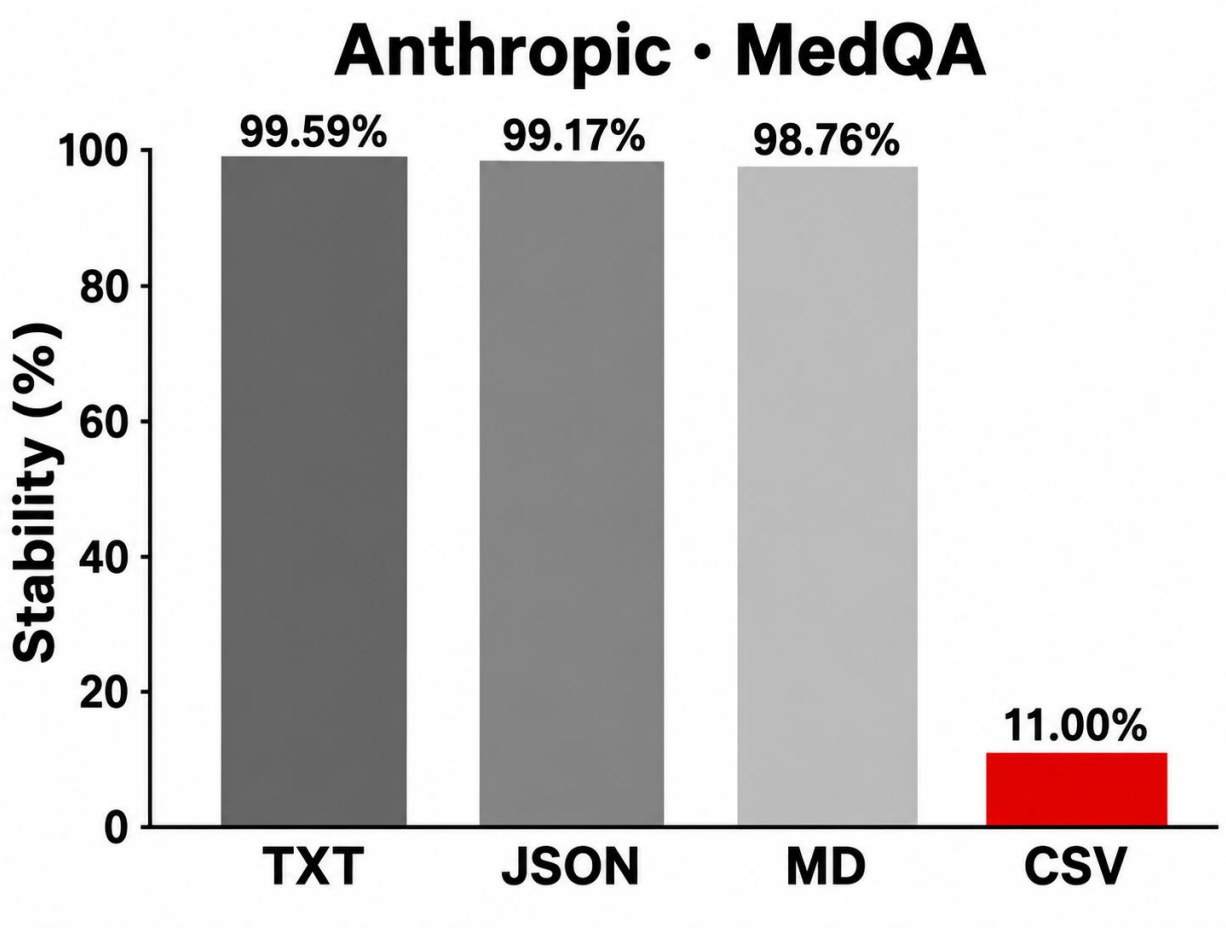
- **Average MR3 violation:**

  45.33%

- **Lowest stability:**

  11.00%

  under CSV (Anthropic workflow)



RQ4

Vote Aggregation

Same format → Repeated runs → Majority vote

Aggregate outputs after inference.

Reactive: fix after inference

VS

Format Routing

Input format → Format router → Optimal format → Workflow

Convert to an empirically optimal format before inference.

Proactive: fix before infrence

RQ4

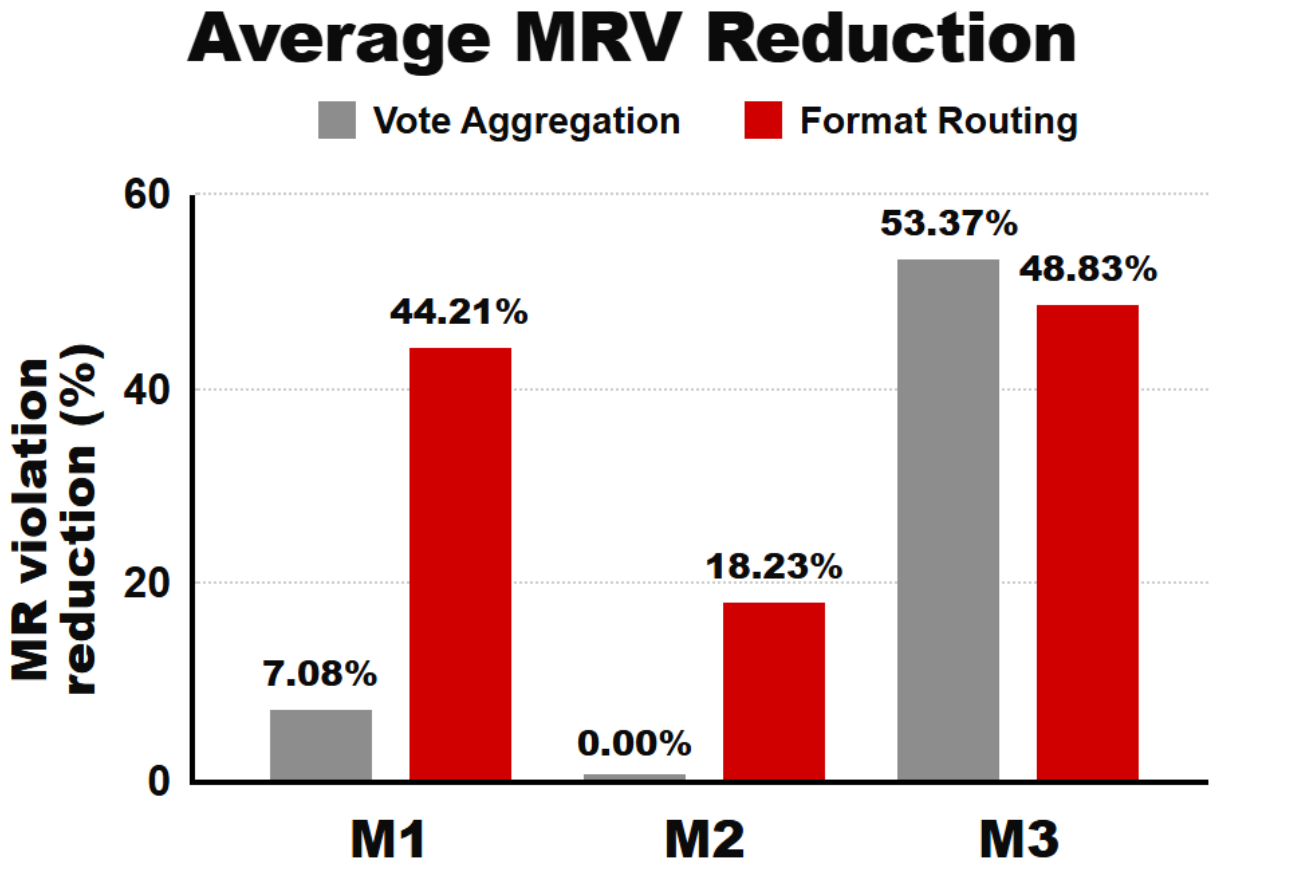
Format routing mitigates format-induced errors.

- **OpenAI–Construct accuracy:**

  +28.74%

- **Vote: M_2 reduction:**

  0.00%



Conclusion

- The **first format-aware metamorphic testing framework** for LLM document workflows.
- A **large-scale empirical study** across four workflows and four formats.
- A **format routing strategy** to mitigate format-induced errors without model retraining.

Thank you for listening!

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Our Repository

Paper